

Computer Practical

Shallow-Water Model

Waves, rotation, instability and large-scale flow

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1 Overview

In this practical you will use a two-dimensional shallow-water model to investigate how a simple set of conservation equations can reproduce several important features of geophysical fluid flow. You will begin with gravity waves, then examine the effect of changing fluid depth, barotropic instability and Rossby waves forced by topography. Finally, you will use a pressure field derived from reanalysis data as the initial state for a very simple weather forecast.

The purpose is not to treat the shallow-water equations as a complete model of the atmosphere or ocean. Instead, use each controlled experiment to connect a visible feature of the simulation with a physical mechanism represented by the equations.

Keep a practical record: save each animation using a meaningful filename and write one or two sentences describing the settings and the main result.

By the end of the practical you should be able to:

- relate the propagation speed of a shallow-water gravity wave to $c = \sqrt{gh}$;
- explain why a gravity wave changes as the fluid becomes shallower;
- explain why an unstable balanced flow requires a perturbation before the instability becomes visible;
- describe the roles of rotation, potential-vorticity conservation and the β effect in large-scale waves;
- recognise both the usefulness and the limitations of a shallow-water model as an atmospheric model;
- recognise a numerical-stability failure caused by an excessively large timestep.

2 What the model represents

The model contains a single layer of fluid with depth $h(x, y, t)$ over fixed orography of height $H(x, y)$. The horizontal wind components are u and v . The height of the upper surface is therefore $h + H$ (Figure 1). The model domain is periodic in the zonal (x) direction.

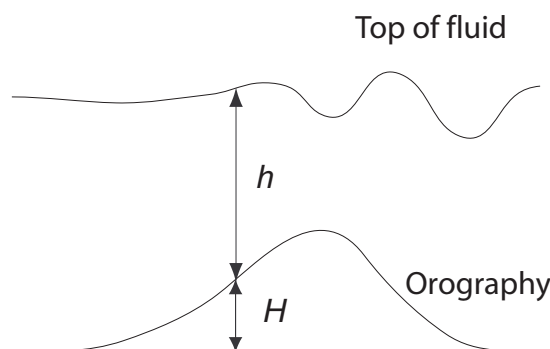


Figure 1: Schematic of the shallow-water model. The prognostic fluid depth is h and the fixed lower-boundary height is H ; the plotted upper-surface height is $h + H$.

In the default configuration, the horizontal grid spacing is 100 km and the domain is intended to resemble a mid-latitude channel around the Earth. A circumference at 45° latitude is about $2\pi R_E \cos 45^\circ \approx 28,000$ km, comparable with the zonal size of the model domain.

The Coriolis parameter varies linearly with y :

$$f(y) = f_0 + \beta(y - \bar{y}), \quad (1)$$

so $f = f_0$ at the centre of the domain. Relative vorticity is

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (2)$$

The lower panel of each animation shows ζ ; the upper panel shows the upper-surface height together with wind vectors.

For atmospheric experiments, the upper-surface height can be interpreted as a pressure-like or geopotential-like field. In the supplied reanalysis experiment the conversion is chosen so that a height of roughly 10 km corresponds to a surface pressure of order 1000 hPa. This is an analogy: the model still contains only one shallow-water layer and has no explicit vertical atmospheric structure.

3 Model equations and numerical method

A two-dimensional conservation equation for a quantity q can be written

$$\frac{\partial q}{\partial t} + \frac{\partial(uq + U)}{\partial x} + \frac{\partial(vq + V)}{\partial y} = Q, \quad (3)$$

where Q is a source term and U and V represent additional flux terms. In the shallow-water model the conserved quantities are fluid volume and horizontal momentum, so the prognostic variables are h , uh and vh :

$$\frac{\partial h}{\partial t} + \frac{\partial(uh)}{\partial x} + \frac{\partial(vh)}{\partial y} = 0, \quad (4)$$

$$\frac{\partial(uh)}{\partial t} + \frac{\partial(u^2h + gh^2/2)}{\partial x} + \frac{\partial(uvh)}{\partial y} = h \left(fv - g \frac{\partial H}{\partial x} \right), \quad (5)$$

$$\frac{\partial(vh)}{\partial t} + \frac{\partial(uvh)}{\partial x} + \frac{\partial(v^2h + gh^2/2)}{\partial y} = h \left(-fu - g \frac{\partial H}{\partial y} \right). \quad (6)$$

The model advances these equations using a two-step Lax–Wendroff method. For the generic equation (3), the first step estimates values half a grid box and half a timestep away. For example,

$$q_{i+1/2,j}^{n+1/2} = \frac{q_{i,j}^n + q_{i+1,j}^n}{2} - \frac{\Delta t}{2\Delta x} \left[(uq + U)_{i+1,j}^n - (uq + U)_{i,j}^n \right], \quad (7)$$

$$q_{i,j+1/2}^{n+1/2} = \frac{q_{i,j}^n + q_{i,j+1}^n}{2} - \frac{\Delta t}{2\Delta y} \left[(vq + V)_{i,j+1}^n - (vq + V)_{i,j}^n \right]. \quad (8)$$

The second step uses the midpoint fluxes to update the grid point:

$$q_{i,j}^{n+1} = q_{i,j}^n + \Delta t \left[- \frac{(uq + U)_{i+1/2,j}^{n+1/2} - (uq + U)_{i-1/2,j}^{n+1/2}}{\Delta x} - \frac{(vq + V)_{i,j+1/2}^{n+1/2} - (vq + V)_{i,j-1/2}^{n+1/2}}{\Delta y} + Q_{i,j}^n \right]. \quad (9)$$

Applied to h , uh and vh , this gives a second-order scheme while retaining the conservative form of the equations. The Coriolis terms are advanced with a centred Crank–Nicolson update. For a pure inertial oscillation this acts as a rotation of the horizontal momentum and therefore does not spuriously increase the wind speed.

Like any explicit numerical scheme, Lax–Wendroff has a stability limit. A useful guide is the Courant number: information should not travel across too many grid boxes in one timestep. The code prints an approximate initial two-dimensional Courant number at the start of every run and warns if it is too large.

4 Log in and download the code

Use the same SSH/SFTP workflow as in *Tools of the Trade*. On campus, the teaching-server address is 130.88.66.57; otherwise use the server details supplied for the course. Replace YOUR_USERNAME with your own username.

Keep two terminal windows open. The **SSH window** is your command line on the teaching server: use it to edit files and run the model. The **SFTP window** is used to copy finished animations from the server back to your own computer.

In the SSH window:

```
ssh -Y YOUR_USERNAME@130.88.66.57
```

In a second, local terminal window:

```
sftp YOUR_USERNAME@130.88.66.57
```

Your password will not be displayed while you type it.

In the SSH window, obtain the model:

```
git clone https://github.com/EnvModelling/shallow-water-model-practical
cd shallow-water-model-practical
```

There is no compilation step: this practical uses the Python version of the model.

File-transfer reminder. In SFTP, lpwd shows the folder on *your own computer* into which files will be downloaded. Use lcd to change it if required, for example:

```
lpwd
lcd Desktop
```

The general form of the download command is

```
get REMOTE_FILE LOCAL_FILENAME
```

The second filename is the name of the copy on your own computer. Giving each result a different local filename lets you keep several experiments even though the model always uses the same filename on the server.

5 Inspecting and running the model

The settings you change are separated from the numerical code. Open the configuration file with line numbers displayed in the SSH window:

```
nano -l python/config.py
```

The main parameters used below are grouped around lines 26–38. Approximate line numbers are useful with nano -l, but they can change when the repository is updated. If a parameter is not where expected, use Ctrl-W and search for its name.

Run the model from the repository root in the SSH window:

```
python3 python/shallow_water_model.py
```

The model prints its progress and, when the animation has been created, tells you where it was saved. The output file on the server is always

```
/tmp/YOUR_USERNAME/animation.gif
```

A later run will overwrite this server copy, so download an animation before starting the next experiment. In the SFTP window, give the local copy a useful name at the same time. For example:

```
get /tmp/YOUR_USERNAME/animation.gif gravity.gif
```

This leaves the server file called `animation.gif`, but saves the copy on your own computer as `gravity.gif`. Use the suggested local filename for each later experiment so that you keep all of your results. When the instructions below say “save as `gravity.gif`”, this is the SFTP step they mean.

To restore the supplied configuration before a new experiment, use this in the SSH window:

```
git restore python/config.py
```

This discards edits to the configuration file, so save any settings you want to keep in your notes first.

6 Parameters used in the practical

Table 1: Main settings in `python/config.py`. The line numbers are approximate.

Parameter	Approx. line	Supplied value	Meaning
<code>f</code>	31	$1.0\text{e-}4 \text{ s}^{-1}$	Coriolis parameter at the centre of the domain
<code>beta</code>	32	$1.6\text{e-}11 \text{ m}^{-1} \text{ s}^{-1}$	Meridional gradient of the Coriolis parameter
<code>dt_mins</code>	34	1.0 min	Numerical timestep
<code>output_interval_mins</code>	35	60 min	Time between animation frames
<code>forecast_length_days</code>	36	4 days	Total model integration time
<code>orography</code>	38	FLAT	Fixed lower boundary H
<code>initial_conditions</code>	39	GAUSSIAN_BLOB	Initial upper-surface-height pattern
<code>initially_geostrophic</code>	40	False	Derive winds from geostrophic balance for height-defined initial states
<code>add_random_height_noise</code>	41	False	Add a small perturbation to the initial height field
<code>random_seed</code>	43	1	Makes random perturbations reproducible between runs

The available named choices include:

- `orography`: FLAT, SLOPE, GAUSSIAN_MOUNTAIN, EARTH_OROGRAPHY and SEA_MOUNT;
- `core initial states`: GAUSSIAN_BLOB, ZONAL_JET, UNIFORM_WESTERLY and REANALYSIS;
- `further initial states`: SHARP_SHEAR, SINUSOIDAL and EQUATORIAL_EASTERLY.

The UNIFORM_WESTERLY state is constructed from a free-surface slope and then the wind is diagnosed from geostrophic balance. Keep this detail in mind for the Rossby-wave experiment below.

Note. Keep `nx=254` and `ny=50` when using EARTH_OROGRAPHY or REANALYSIS, because the supplied data files use that grid.

7 Experiments

For each subsection, restore `python/config.py` before entering the listed settings unless the instructions explicitly say to modify the preceding run. Save the animation before starting the next run.

7.1 Gravity waves and numerical stability

A non-uniform free surface with the winds initially at rest is not in force balance. The resulting pressure-gradient force launches gravity waves. Remove rotation so that the response is as simple as possible. Set:

```
f = 0.0
beta = 0.0
orography = FLAT
initial_conditions = GAUSSIAN_BLOB
initially_geostrophic = False
add_random_height_noise = False
```

```
output_interval_mins = 15.0
forecast_length_days = 0.5
```

Run the model and save the animation as, for example, `gravity.gif`.

Prediction. For a layer depth of about 10 km, estimate $c = \sqrt{gh}$. Roughly how far should a gravity wave travel in one hour? Use the horizontal axis of the animation to make a prediction before viewing the result.

Exercise. Estimate the propagation speed from the animation and compare it with $\sqrt{gh}$. Is there much signal in the relative-vorticity panel? Why are gravity waves different in this respect from rotational waves?

Optional numerical-stability test. Keep the gravity-wave settings but change only

```
dt_mins = 5.0
```

The output interval of 15 minutes is still an exact multiple of the timestep. Run the model again. The printed Courant number is now greater than one and the integration should become unstable; the updated code stops with a clear error once it produces an invalid fluid depth.

Exercise. What has changed physically between the 1-minute and 5-minute experiments? What has changed numerically? Why does a numerical-stability failure not represent a real atmospheric or oceanic process?

7.2 A tsunami analogue: changing fluid depth

A tsunami is also a shallow-water gravity wave, but its speed depends on the local ocean depth. Restore the supplied configuration, then use the gravity-wave setup with a submerged Gaussian seamount:

```
f = 0.0
beta = 0.0
orography = SEA_MOUNT
initial_conditions = GAUSSIAN_BLOB
initially_geostrophic = False
add_random_height_noise = False
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run and save the result as `tsunami.gif`.

Prediction. Since $c = \sqrt{gh}$, what should happen to the wave speed as the wave enters shallower water over the seamount? If the period remains similar, what should happen to its wavelength?

Exercise. Identify the changes in the wave as it encounters the shallower region. Which features resemble tsunami shoaling? Remember that the model uses a highly idealised topography and a much larger surface disturbance than a real tsunami.

7.3 Barotropic instability

Large-scale flows can contain strong horizontal shear. In a shallow-water system, a necessary condition for barotropic instability is that the meridional gradient of potential vorticity,

$$\frac{\partial}{\partial y} \left(\frac{f + \zeta}{h} \right), \quad (10)$$

changes sign somewhere in the flow. Start with an idealised Bickley-like jet:

```
f = 1.0e-4
beta = 1.6e-11
orography = FLAT
initial_conditions = ZONAL_JET
initially_geostrophic = True
add_random_height_noise = True
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run and save the result as `jet_noise.gif`. The fixed `random_seed` means that all students should apply the same small perturbation.

Now change only

```
add_random_height_noise = False
```

and save the result as `jet_no_noise.gif`.

Prediction. An instability amplifies perturbations, but does it create a perturbation from an exactly symmetric state? Predict which run should develop visible waves first.

Exercise. Compare the two animations. Why does the random perturbation matter even though the underlying jet is unstable? Relate your answer to the distinction between an unstable basic state and the disturbance that grows on it.

Note. If time permits, repeat the noisy experiment with `initial_conditions = SHARP_SHEAR`. This produces a more frontal, cyclogenesis-like example of shear instability.

7.4 Orographic Rossby waves

A dominant feature of the Earth's atmosphere is the presence of large-scale Rossby, or planetary, waves. In the Northern Hemisphere these are forced in part by flow over major mountain ranges such as the Himalayas and Rockies. In a rotating shallow-water system, motion over topography changes the fluid depth and therefore the relative vorticity required to conserve potential vorticity. The β effect provides an additional planetary-vorticity restoring mechanism.

Start with an isolated Gaussian mountain. Set

```
f = 1.0e-4
beta = 1.6e-11
orography = GAUSSIAN_MOUNTAIN
initial_conditions = UNIFORM_WESTERLY
initially_geostrophic = True
add_random_height_noise = True
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run the model and save the animation as, for example, `mountain_beta.gif`.

Exercise. Explain the initial meander in terms of vortex stretching and compression as the fluid passes over the mountain. Where is relative vorticity generated, and why?

A deliberately unsuccessful experiment. Now change only

```
f = 0.0
beta = 0.0
```

and run the model again. Do not try to “fix” the result immediately. Inspect what Python prints in the terminal.

Exercise. Why do NaN values appear? In `python/shallow_water_model.py`, find the section beginning `if cfg.initially_geostrophic`. Look carefully at the expressions used to calculate u and v . Which quantity appears in a denominator, and what value does it have after you set $f = 0.0$ and $\beta = 0.0$? Explain why the initial state cannot be constructed. Is this the physical response of a non-rotating 20 m s^{-1} westerly, or has the problem occurred before the dynamical experiment can begin?

Note. This is a useful modelling lesson: changing a physical parameter can invalidate an assumption used to construct the initial state. A failed run can therefore tell you something about the model formulation rather than about the subsequent dynamics.

Finally, restore

```
f = 1.0e-4
beta = 0.0
```

with the other mountain settings unchanged. Run the model and save it as `mountain_fplane.gif`.

Exercise. Compare the f -plane run with your original β -plane run. What happens to the disturbance after the air passes over the mountain? Why is the response different when f has no meridional gradient? Think particularly about the restoring mechanism for a Rossby wave.

Note. If time permits, restore the β -plane values and replace `GAUSSIAN_MOUNTAIN` with `EARTH_OROGRAPHY`. What recognisable features of large-scale Northern Hemisphere flow emerge from realistic topography?

7.5 A simple weather forecast

The final core experiment initializes the model with a pressure field derived from reanalysis data from 1 July 2000. The original field is from the Southern Hemisphere and has been transformed for use in this Northern Hemisphere teaching configuration. Restore the supplied configuration and set:

```
f = 1.0e-4
beta = 1.6e-11
orography = FLAT
initial_conditions = REANALYSIS
initially_geostrophic = True
add_random_height_noise = False
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run and save the result as `weather.gif`.

Exercise. At the start of the forecast, why do some relatively rapid gravity-wave-like adjustments appear? What imbalance in an imperfectly initialized shallow-water state could launch them? As the run develops, which aspects look meteorological and which remind you that this is only a one-layer model?

7.6 Further experiments if time permits

The model contains several additional initial conditions that are useful for exploration after the core comparisons above. Restore the supplied configuration before each one, then change only the settings listed.

Jupiter-like alternating jets

```
initial_conditions = SINUSOIDAL
initially_geostrophic = True
add_random_height_noise = True
```

The alternating zonal jets can generate vortices that interact and merge, illustrating the tendency of two-dimensional/geostrophic turbulence to transfer energy towards larger scales. How do the long-lived vortices compare qualitatively with Jupiter's Great Red Spot?

Equatorially trapped waves

```
f = 0.0
beta = 2.5e-11
initial_conditions = EQUATORIAL_EASTERLY
initially_geostrophic = True
add_random_height_noise = True
output_interval_mins = 120.0
forecast_length_days = 8.0
```

Because f changes sign across the middle of the domain, disturbances can become trapped near the model equator.

Equatorial Kelvin wave

```
f = 0.0
```

```
beta = 5.0e-10  
output_interval_mins = 15.0  
forecast_length_days = 0.5
```

Compare the eastward and westward responses. Which disturbance remains trapped near the equator, and how does rotation make eastward and westward propagation asymmetric?

8 What you should take away

A very small set of conservation laws can reproduce a surprisingly wide range of behaviour. Gravity provides fast restoring waves; changing fluid depth changes their speed. Rotation links horizontal pressure gradients to balanced flow and makes potential-vorticity conservation dynamically important. Sheared balanced flows can amplify perturbations, while topography and the meridional change of planetary vorticity generate large-scale wave responses.

The same practical also illustrates two modelling lessons. First, a model experiment is easiest to interpret when one physical assumption is changed at a time and all other settings are controlled. Second, a failed or strange run is not automatically a physical result: timestep choice and initialization both matter. The $f = 0$ westerly experiment is a particularly clear example because the geostrophic initial condition itself becomes mathematically undefined. A useful model user therefore asks both “what process is represented?” and “could this feature have been produced by the numerical method or the experimental setup?”